The Last Degree
The Graduate ROI Intelligence System

Manav Patel **Dhruv Patel** **Drashti Shah** **Parth Patel**

Department of Applied Data Intelligence
San Jose State University
San Jose
CA, USA

Abstract—This paper details the design, implementation, and analytical architecture of **The Graduate ROI Intelligence System** (The Last Degree), a cloud-native data warehouse solution providing real-time, risk-adjusted intelligence on the Return on Investment (ROI) of higher education degrees. The system addresses the increasing career uncertainty driven by escalating student debt and AI-driven job market disruption by integrating four crucial data domains: University Performance (College Scorecard), Live Job Market (Adzuna), Layoff Events (WARN Notices), and Economic Trends (BLS). The architecture employs a robust ELT (Extract, Load, Transform) paradigm, utilizing **Apache Airflow** for orchestration, **Snowflake** as the scalable cloud data warehouse, and **dbt (data build tool)** for complex, version-controlled business logic transformations. The final output is the dynamic **"Degree ROI Score,"** materialized into the mart_degree_roi_and_industry_outlook analytical mart, which powers a responsive, user-facing application built with React and featuring an integrated SQL AI Agent.

Keywords —Data Warehousing, ELT, Apache Airflow, dbt, Snowflake, Real-time Analytics, Degree ROI, Job Market Intelligence, Data Marts, SQL AI Agent.

I. INTRODUCTION

The value proposition of advanced degrees is facing unprecedented scrutiny. As stated by the CEO of Anthropic, it is forecast that "50% of entry-level white-collar jobs will be gone within five years" due to AI automation. Coupled with the common burden of over $150,000 in student debt for graduate programs, the question, "Relevance. Return. Reality: what will your degree be worth in five years?" has become paramount.

Traditional educational planning relies on static, historical data, such as past earnings averages, which inherently fail to capture real-time market risk, job velocity, or the immediate threat of mass layoffs.

The project, The Graduate ROI Intelligence System (The Last Degree), resolves this deficit by building an end-to-end data platform that moves beyond historical performance to quantify and forecast the Return on Investment (ROI) of graduate programs. The system integrates historical educational outcomes with real-time labor market signals to deliver a dynamic, risk-adjusted analytical product.

II. PROBLEM STATEMENT

The core technical and analytical challenge lies in the **fragmentation and heterogeneity** of the data required for a holistic ROI assessment. The system must overcome three key obstacles:

1. **Siloed Data Integration:** Critical factors University Performance, Live Job Market demand, WARN Act layoff risk, and BLS employment projections exist in separate formats (APIs) and must be unified via a common taxonomy for meaningful cross-analysis.

2. **Lack of Transformation Governance:** The complex business logic needed to calculate the multi-factor Degree ROI Score must be housed in a traceable, testable, and version-controlled environment, which is the mandate of the dbt layer.

3. **Real-Time Data Velocity:** The system must handle the high-velocity, daily ingestion of transactional data (e.g., 22,500 job listings per run) and structure it for low-latency querying by the user-facing application.

III. DATA SOURCES AND DETAILS

The analytical power of the system derives from the successful integration of four distinct data pillars, each contributing a vital component to the final Degree ROI Score.

Data Domain	Source API/Feed	Data Velocity	Key Analytical Contribution	Output Table (RAW Schema)
University Performance	U.S. College Scorecard API	Annually	Historical cost, debt, and median earnings (10 years post-graduation).	COLLEGE_SCORECARD_DATA
Live Job Market	Adzuna JobAPI	Daily (2 AM UTC)	Real-time job posting velocity, current salary distributions, and skill demand.	JOB_LISTINGS
Economic Trends	U.S. Bureau of Labor Statistics (BLS) API	Monthly	Macro-level context, 10-year employment projections, and median wages.	BLS_EMPLOYMENT_PROJECTIONS
Layoff/Job Cuts	CA WARN Notices API	Monthly	Real-time risk signal (career stability index) for specific companies and sectors.	WARN_EVENTS

College Scorecard Data: This core dataset is pulled daily via the Department of Education's robust API. It provides institution-level performance metrics, including average net price, student debt loads, and median earnings 10 years after entry. The ETL phase must handle flattening deeply nested JSON structures and managing pagination across thousands of institutions (up to 500 pages) using exponential backoff to handle API rate limiting.

Adzuna Job Listings: Retrieved daily via the Adzuna Job API, this high-velocity dataset captures real-time market demand. Each run fetches approximately 22,500 jobs across 30 categories. The ETL phase focuses on cleaning HTML from job descriptions, normalizing variable salary strings to a numeric format, and handling API rate limits while fetching data across 15 pages per category.

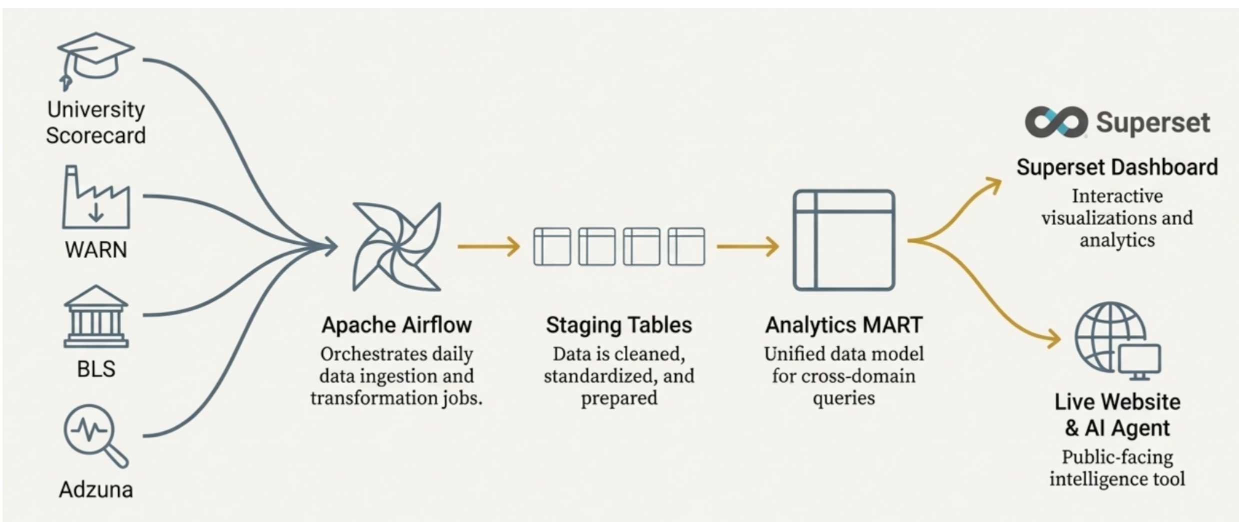
BLS Employment Projections: This monthly data is accessed via the U.S. Bureau of Labor Statistics API. It provides occupational employment statistics. The essential ETL phase involves a complex transformation: splitting approximately 834 macro-occupations into over 5,000 granular job title rows and performing rigorous data type conversion to prepare the structured data for the data warehouse.

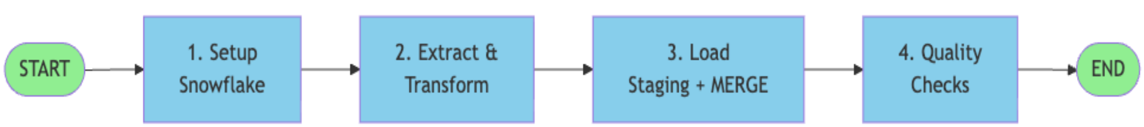
WARN Events: The Worker Adjustment and Retraining Notification (WARN) Act notices are accessed monthly via a dedicated API. This dataset records mass layoffs and closures. The ETL phase focuses on cleaning location strings and validating and converting worker count strings to numeric values, which is necessary before calculating the Layoff Risk Index.

IV. METHODOLOGY AND SYSTEM ARCHITECTURE

The system is built upon a data processing architecture leveraging the best attributes of both ETL and ELT within a cloud-native environment.

Component	Technology	Role and Function
Orchestration	Apache Airflow (Docker)	Manages all DAGs, connections, and complex job dependencies.
Data Warehouse	Snowflake	The scalable cloud repository for all data: **RAW** (ingestion), **ANALYTICS** (marts), and **SNAPSHOTS** (history).
Transformation	dbt (data build tool)	Governs the complex **ELT** layer, running tests, documentation, and the centralized business logic.
Presentation	React / Superset	Provides the final, high-speed interfaces for end-user consumption.





A. Airflow Pipeline Orchestration

Apache Airflow serves as the central control plane, orchestrating the data ingestion processes. The pipelines are designed to handle both data volume (Adzuna) and complex file parsing (BLS).

Each pipeline is configured with a dedicated DAG, employing a **staging table + MERGE** pattern for reliable Upsert operations into the Snowflake RAW schema.

Pipeline	**ETL Steps Performed in Airflow**	**ELT Steps Performed in dbt (Snowflake)**
BLS ETL	Parsing CSV, splitting ~834 macro-occupations into ~5,000+ granular rows, handling NULL values.	Joining BLS projections with institution ROI based on occupation groups.
Adzuna ETL	Cleaning HTML content, normalizing salary strings, handling API pagination/retries, location parsing.	Aggregating job posting counts, calculating median salaries by occupation group.
College Scorecard ETL	Flattening nested JSON, type casting, handling API exponential backoff.	Calculating ROI and Value Index and joining with market/risk data.
WARN ETL	Reading CSV, cleaning location strings, validating worker count strings.	Calculating the **Layoff Risk Index** and integrating it with the final mart.

B. dbt Transformation Methodology

The dbt project is the analytical core, managing the **ELT** logic within Snowflake, where all complex joins and business rules reside.

1) Transformation Layers and Metric Computation

- **Staging Layer (stg_* Models):** Materialized as Views in the RAW schema. These models perform final data preparation and initial metric calculation (e.g. ROI_IN_STATE).
- **Mart Layer (mart_* Models):** The core analytical output is materialized as the **mart_degree_roi_and_industry_outlook** Table in the ANALYTICS schema for high-

speed querying. It is created by a cross-join of all institutions with the 25 occupation groups, linking financial data with market intelligence.

2) The Degree ROI Score Calculation

The dynamic **Degree ROI Score** is the central composite metric, integrating financial return with market stability metrics:

*ROI Score = [Normalized Earnings/Normalized Cost] * Weighted Demand * (1 - Layoff Risk Index)*

The **Layoff Risk Index** (derived from the ratio of WARN layoff counts to active Adzuna job postings for a specific occupation group) is a crucial, real-time measure of career stability.

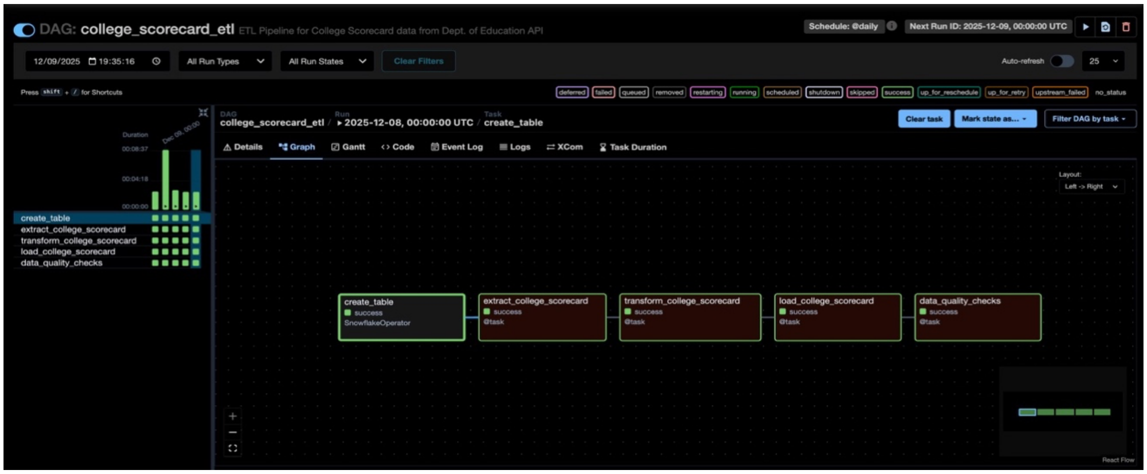
C. ETL+ELT Airflow Pipeline

1. College Scorecard ETL Pipeline

This pipeline provides the fundamental data for the financial return component of the Degree ROI Score.

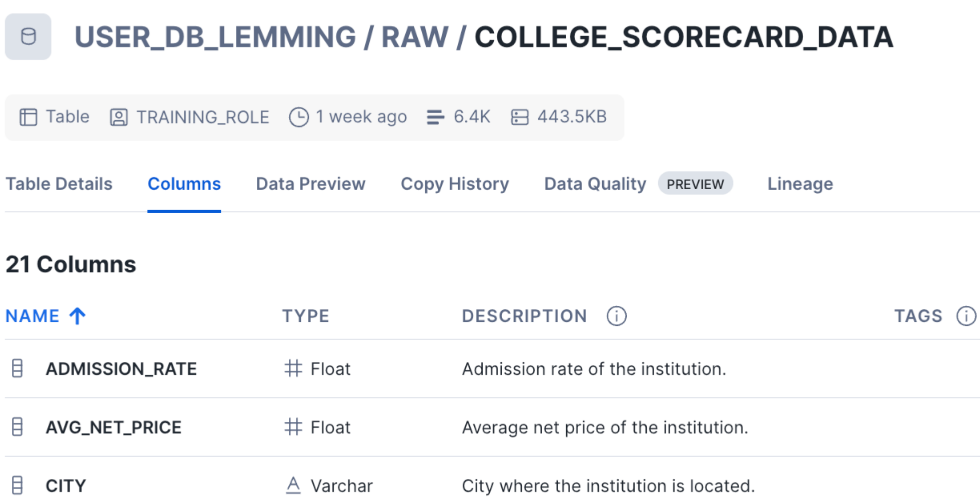
Feature	**Details**
Source	U.S. Department of Education College Scorecard **API**
Data Velocity	Annually
Airflow ETL Focus	The API returns nested JSON data across multiple pages (~500 pages for ~50,000 institutions). The ETL process is critical for: **JSON Flattening:** Un-nesting complex fields (e.g. earnings data) to create flat, relational records. **Pagination & Error Handling:** Implementing loops with **exponential backoff** to handle API rate limiting and process large volumes of institutional data reliably. **Type Casting:** Converting raw data strings (e.g., tuition, debt figures) into correct numeric types.
Snowflake ELT Focus	The dbt models (stg_institution) use this cleaned data to calculate foundational ROI metrics like **ROI_IN_STATE** (Earnings (10 years) In-state tuition) and the **VALUE_INDEX** before joining with market data.
Output	RAW.COLLEGE_SCORECARD_DATA

Airflow:



<u>**Table Schema & Details:**</u>

- **Source:** U.S. Department of Education API.
- **The "What":** This table contains exhaustive records for over 50,000 higher education institutions.
- **Key Columns:** UNITID (Primary Key), INSTNM (Name), COSTT4_A (Annual tuition), DEBT_MDN (Median student debt), and MDN_EARN_WNE_P10 (Median earnings 10 years after entry).
- **Role:** It provides the **financial baseline**. Without this table, the system cannot calculate the "Investment" side of the ROI equation.



USER_DB_LEMMING / RAW / COLLEGE_SCORECARD_DATA

Table | TRAINING_ROLE | 1 week ago | 6.4K | 443.5KB

Table Details | Columns | Data Preview | Copy History | Data Quality PREVIEW | Lineage

21 Columns

NAME	TYPE	DESCRIPTION	TAGS
ADMISSION_RATE	Float	Admission rate of the institution.	
AVG_NET_PRICE	Float	Average net price of the institution.	
CITY	Varchar	City where the institution is located.	

COMPLETION_RATE	Float	Completion rate of the institution.
EARNINGS_10YRS	Float	Ten-year earnings of the institution.
EARNINGS_6YRS	Float	Six-year earnings of the institution.
FEDERAL_GRANTS	Float	Amount of federal grants provided to stu...
INSTITUTION_ID	Number	Unique ID of the institution.
INSTITUTION_NAME	Varchar	Name of the institution.
LOAD_TIMESTAMP	Timestamp...	Timestamp when the data was loaded.
LOANS_AMOUNT	Float	Amount of loans provided to students.
MEDIAN_DEBT	Float	Median debt of students after graduation.
OWNERSHIP	Number	Ownership type of the institution.
REGION_ID	Number	Region ID of the institution.

2. Adzuna Job Listings ETL Pipeline

This pipeline supplies the **real-time job demand** data, which is crucial for the **Weighted Demand** factor and the denominator of the Layoff Risk Index.

Feature	**Details**
Source	Adzuna Job **API**
Data Velocity	Daily (Scheduled for off-peak hours like 2 AM UTC)
Airflow ETL Focus	This is a high-volume, high-frequency pipeline. The ETL process manages: **API Pagination:** Iterating through multiple pages (~15 pages) across multiple categories (~30) to fetch large batches (~22,500 jobs per run). **Data Normalization:** Cleaning unstructured fields, such as stripping HTML tags from job descriptions. **Salary Standardization:** Converting variable salary strings standardized numeric fields (min/max/median) to be usable for analytical modeling.
Snowflake ELT Focus	The dbt models aggregate the cleaned job listings data by location and the **25-occupation-group taxonomy** to compute job posting velocity and median salary trends for the analytical mart.

Feature	Details
Output	RAW.JOB_LISTINGS

Airflow:

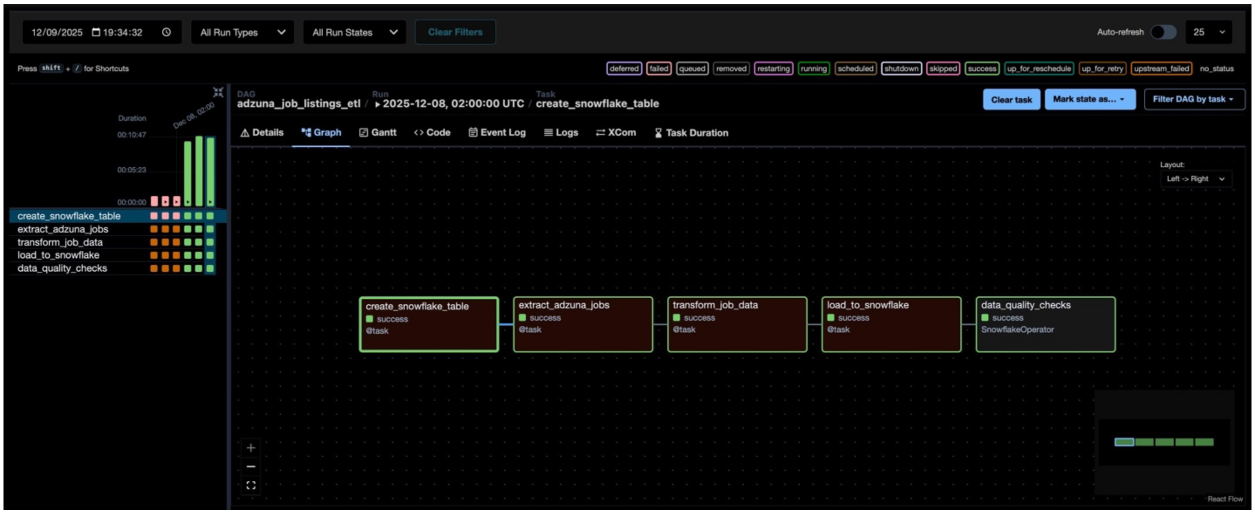
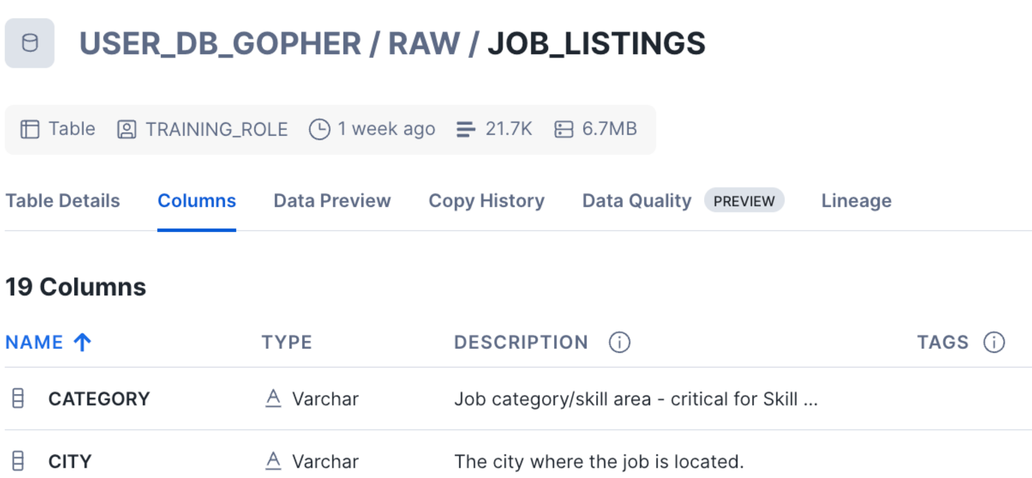


Table Schema & Details:

- **Source:** Adzuna Job Market API.
- **The "What":** A high-velocity table storing approximately 22,500 active job postings per daily run.
- **Key Columns:** ADZUNA_ID, CATEGORY (Adzuna's native grouping), SALARY_MIN/MAX, and CREATED_DATE.
- **Role:** It provides **real-time demand**. This table allows the system to see which careers are hiring *today* versus what the historical data suggests.



COMPANY	Varchar	The company offering the job.
CONTRACT_TIME	Varchar	The type of contract, either part-time or ...
CONTRACT_TYPE	Varchar	The type of contract, either permanent o...
DESCRIPTION	Varchar	A detailed description of the job.
EXTRACTED_AT	Timestamp...	The date and time the job data was extr...
JOB_ID	Varchar	Unique identifier from Adzuna API
JOB_TITLE	Varchar	The title of the job.
LATITUDE	Float	The latitude of the job location.
LOAD_DATE	Date	Date when this record was loaded into S...
LOCATION	Varchar	The location of the job.
LONGITUDE	Float	The longitude of the job location.
POSTING_DATE	Date	The date the job was posted.

3. BLS Employment Projections ETL Pipeline

This pipeline provides the **macroeconomic context** and long-term stability projections necessary for comprehensive ROI assessment.

Feature	Details
Source	U.S. Bureau of Labor Statistics (BLS) **API**
Data Velocity	Monthly
Airflow ETL Focus	This ETL is computationally intensive: **Complex Transformation:** The core task is decomposing ~834 macro-occupations from the source into ~5,000+ highly **granular job title rows**. This structural transformation is performed in the ETL layer to simplify the downstream ELT joining. **Data Structuring:** Ensuring consistent data types and handling missing values across the projection fields (e.g., 10-year growth rates).

Feature	Details
Snowflake ELT Focus	dbt models join this granular, projected job data with the institutional ROI metrics using the **25-occupation-group taxonomy** to provide a forward-looking dimension to the final score.
Output	RAW.BLS_EMPLOYMENT_PROJECTIONS

Airflow:

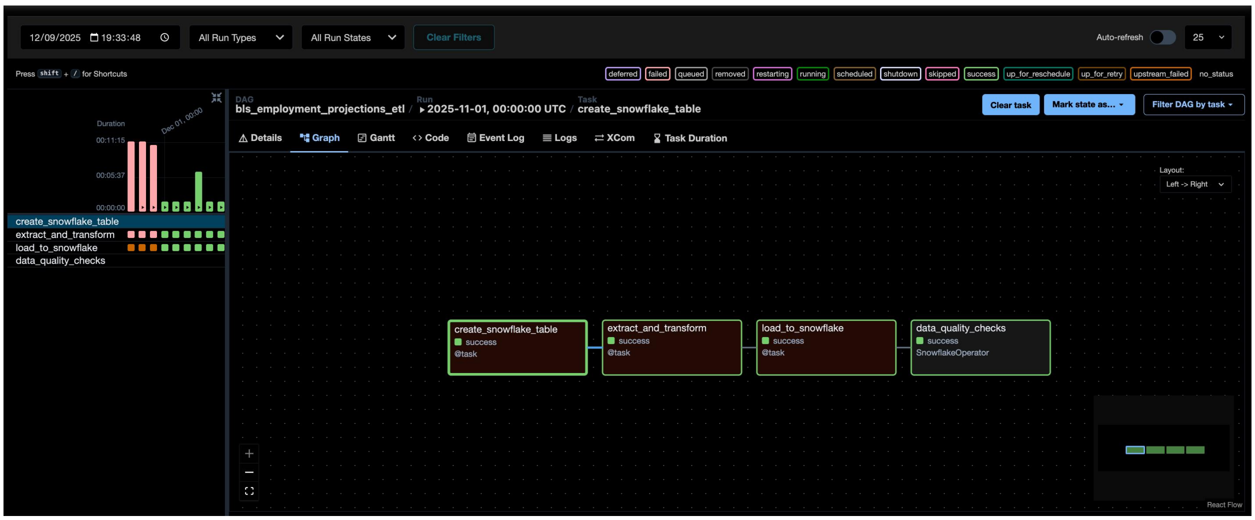
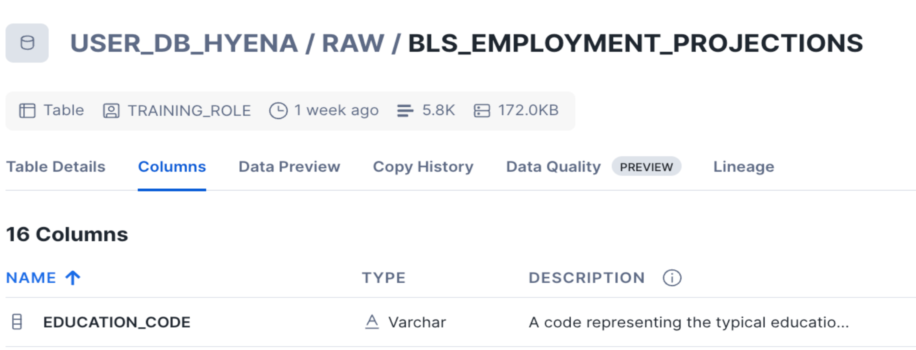


Table Schema and Details:

- **Source:** Bureau of Labor Statistics API/Feed.
- **The "What":** A monthly snapshot of the U.S. labor outlook for the next decade.
- **Key Columns:** OCC_CODE (Standard Occupational Classification), EMPLOYMENT_BASE, EMPLOYMENT_PROJECTION, and GROWTH_PCT.
- **Role:** It provides **long-term stability**. While Adzuna shows today's demand, the BLS data forecasts if a degree would still be relevant in 10 years.



EMPLOYMENT_2024	Varchar	The number of people employed in a giv...
EMPLOYMENT_2034	Varchar	The number of people employed in a giv...
EMPLOYMENT_CHANGE_2024_2...	Varchar	The change in employment from 2024 to...
EMPLOYMENT_PERCENT_CHANG...	Varchar	The percentage change in employment f...
JOB_TITLE	Varchar	The title of the job.
LOAD_TIMESTAMP	Timestamp...	The date and time the data was loaded.
MEDIAN_ANNUAL_WAGE_2024	Varchar	The median annual wage for the occupa...
OCCUPATIONAL_OPENINGS_202...	Varchar	The average annual number of occupati...
OCCUPATION_CODE	Varchar	A code representing the occupation.
OCCUPATION_TITLE	Varchar	The title of the occupation.
TR_CODE	Varchar	A code representing the training require...
TYPICAL_ENTRY_LEVEL_EDUCAT...	Varchar	The typical level of education required f...
TYPICAL_ON_THE_JOB_TRAINING	Varchar	The typical amount of on-the-job trainin...

4. WARN Notices ETL Pipeline

This pipeline provides the essential **real-time risk signal** used to calculate the crucial **Layoff Risk Index**.

Feature	**Details**
Source	WARN Act Notices **API**
Data Velocity	Monthly
Airflow ETL Focus	The ETL focuses on data cleaning and validation: **Data Validation:** Checking for complete and sensible records (e.g. ensuring layoff worker counts are non-zero). **String Cleaning:** Standardizing company names and location fields. **Type Conversion:** Rigorously converting raw worker count strings into clean numeric values for aggregation.
Snowflake ELT Focus	The dbt models aggregate the WARN data by industry/occupation group and then divide it by the Adzuna job counts to compute the **Layoff Risk Index**.
Output	RAW.WARN_EVENTS

Airflow:

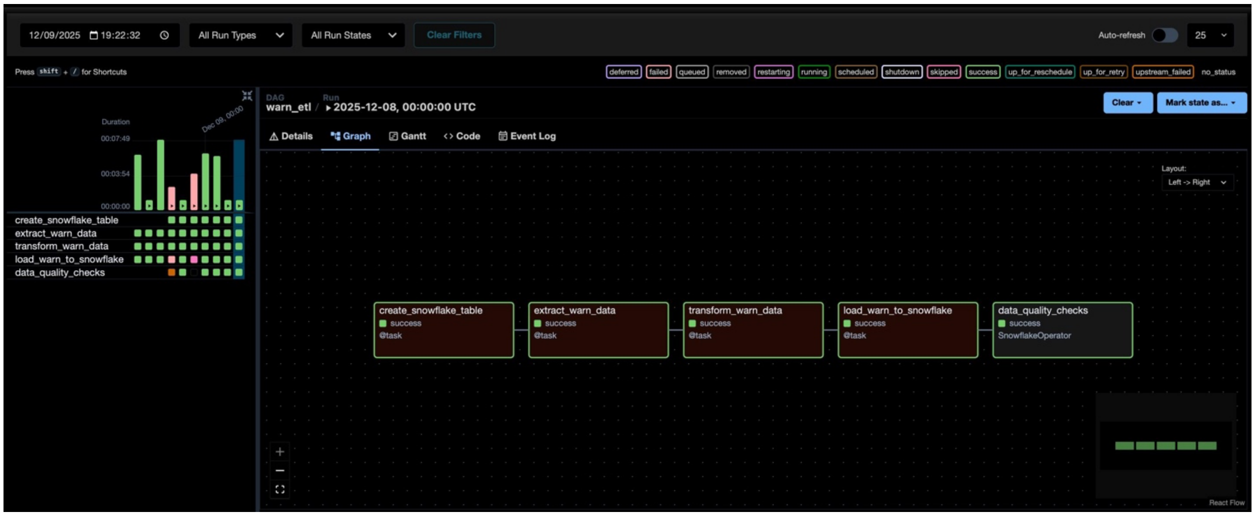


Table Schema & Details:

- **Source:** California WARN Act API/Data Feed.
- **The "What":** A record of legally mandated layoff notices filed by employers.
- **Key Columns:** EVENT_ID, AFFECTED_WORKERS, LAYOFF_DATE, and COMPANY_NAME.
- **Role:** It provides the **risk signal**. This is the most unique part of the system; it acts as a "negative weight" to lower the ROI score of degrees in industries currently experiencing high volatility.

USER_DB_GIRAFFE / RAW / WARN_EVENTS

Table | TRAINING_ROLE | 1 week ago | 3.6K | 135.0KB

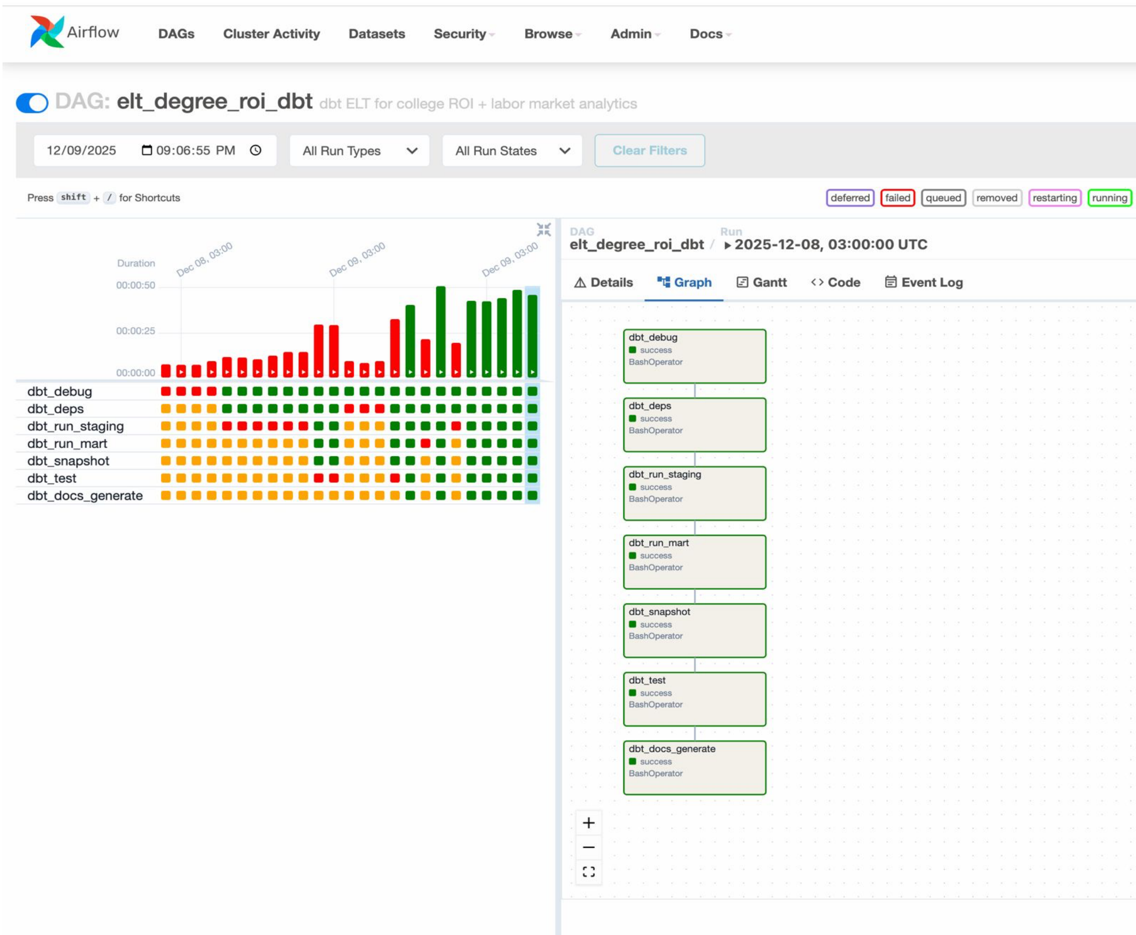
Table Details | Columns | Data Preview | Copy History | Data Quality PREVIEW | Lineage

11 Columns

NAME	TYPE	DESCRIPTION	TAGS
CITY	Varchar	The city where the layoff occurred.	
CLOSURE_LAYOFF	Varchar	The type of layoff, either a closure or a l...	
COMPANY	Varchar	The name of the company that experien...	
COUNTY	Varchar	The county where the layoff occurred.	
EFFECTIVE_DATE	Timestamp...	The date the layoff became effective.	
INDUSTRY	Varchar	The industry of the company that experi...	
LAYOFFDURATION	Varchar	The duration of the layoff, either tempor...	
NUMBER_OF_WORK...	Number	The number of workers affected by the l...	
REGION	Varchar	The region where the layoff occurred.	
STATE	Varchar	The state where the layoff occurred.	
WARN_RECEIVED_D...	Timestamp...	The date the layoff warning was received.	

5. The ELT: Merging Data Sources with dbt in Snowflake

The ELT transformation is structured into two main phases within dbt: the **Staging Layer** and the **Mart Layer**. This modular approach ensures that every step is tested, documented, and fully governed.



A. The Challenge: Data Unification

The four data sources (College Scorecard, Adzuna, BLS, WARN) inherently lack a common primary key. They describe different entities: a university, a job posting, a layoff event, and a macro-economic forecast.

To solve this, the ELT process relies on two key dimensional constructs:

- **Institution ID (UnitID):** Used to link College Scorecard data to the output mart.

- **The 25-Occupation-Group Taxonomy:** This custom, standardized lookup table is the **common language** used to link the economic/market data (Adzuna, BLS, WARN) to the degree programs they enable. Every job, layoff event, and projection must be mapped to one of these 25 groups (e.g., "Software & Data Science," "Finance & Accounting").

B. ELT Staging Layer

- **Input:** The four raw tables from the Airflow ETL process (RAW.COLLEGE_SCORECARD_DATA, etc.).
- **dbt Models:** stg_* models (e.g., stg_institution, stg_job_demand).
- **Goal:** Clean, standardize, and calculate initial metrics within each domain.

Staging Model	**Source Data**	**Initial Transformation / Metric**
stg_institution	College Scorecard	Calculate ROI base : Normalized Earnings / Normalized Cost.
stg_job_demand	Adzuna Job Listings	Aggregate job counts by occupation group, calculate median salary by group.
stg_layoff_risk	WARN Notices	Calculate total affected workers per occupation group per month.
stg_bls_projections	BLS Projections	Map the ~ 5,000 granular job titles to the **25-Occupation-Group Taxonomy**.

C. ELT Mart Layer: Data Merging and Final Calculation

- **Input:** The four clean staging models (stg_*).
- **dbt Model:** mart_degree_roi_and_industry_outlook (The final analytical table).
- **Goal:** Perform multi-way joins and execute the core business logic to produce the final **Degree ROI Score**.

This process follows three critical steps :

I. Cross-Domain Join (The Merge)

The final mart is created by joining the two main dimensions: Institutions and Occupation Groups.

1. A base table is created by taking all unique **Institution IDs** and cross-joining them with all **25 Occupation Groups**. This ensures every institution has an entry for every possible career path.
2. The ROI metrics (stg_institution) are joined onto this base table via **Institution ID**.
3. The market intelligence metrics (Weighted Demand, Layoff Risk, BLS Projection) are joined onto this base table via the **25-Occupation-Group Taxonomy**.

This single mart now contains everything needed for the final score: financial history (Cost/Earnings) and real-time market risk (Demand/Layoffs).

II. Layoff Risk Index Calculation

The heart of the risk-adjustment is the **Layoff Risk Index**. This metric is calculated by comparing negative market signals (Layoffs) against positive signals (Job Demand) for a given occupation group:

$$\text{Layoff Risk Index} = \frac{\text{Aggregated WARN Layoffs}}{\text{Aggregated Adzuna Job Postings}}$$

This is a real-time stability measure. For example, an index of 0.05 means that for every 100 job postings in that field, there were 5 worker layoffs reported.

III. Final ROI Score Calculation

The final materialized table (mart_degree_roi_and_industry_outlook) is clean, complete, and highly performant, ready for direct querying by the SQL AI Agent and Superset dashboards.

IV. TABLE SCHEMA (ANALYTICS)

The final analytical layer is the mart_degree_roi_and_industry_outlook.

- **Materialization**: Table.
- **Purpose**: Final mart that combines all data sources, created via a cross-join of institutions with all occupation groups.
- **Key Metrics**: Includes Institution ROI metrics, job posting counts, layoff risk ratios (layoff-to-posting ratio), and BLS employment projections.

USER_DB_GOPHER / _ANALYTICS / **MART_DEGREE_ROI_AND_INDUSTRY_OUTLOOK**

Table · TRAINING_ROLE · 6 days ago · 160.7K · 1.6MB

Table Details | Columns | Data Preview | Copy History | Data Quality PREVIEW | Lineage

16 Columns

Searc

AVG_JOB_SALARY	Number	Average job salary.
DEBT_TO_EARNINGS_RATIO	Float	Debt to earnings ratio.
EMPLOYEES_LAID_OFF	Number	Number of employees laid off.
EMPLOYMENT_2024	Number	Number of employees employed in 2024.
EMPLOYMENT_2034	Number	Number of employees employed in 2034.
EMPLOYMENT_PCT_CHAN...	Number	Percentage change in employment.
INSTITUTION_ID	Number	Unique identifier for the institution.
INSTITUTION_NAME	Varchar	Name of the institution.
INSTITUTION_STATE	Varchar	State of the institution.
JOB_POSTINGS	Number	Number of job postings.
LAYOFF_TO_POSTING_RATIO	Float	Ratio of layoffs to job postings.
MEDIAN_ANNUAL_WAGE_2...	Number	Median annual wage in 2024.
OCCUPATION_GROUP	Varchar	Occupation group.
ROI_IN_STATE	Float	Return on investment in the state.
ROI_OUT_OF_STATE	Float	Return on investment out of state.

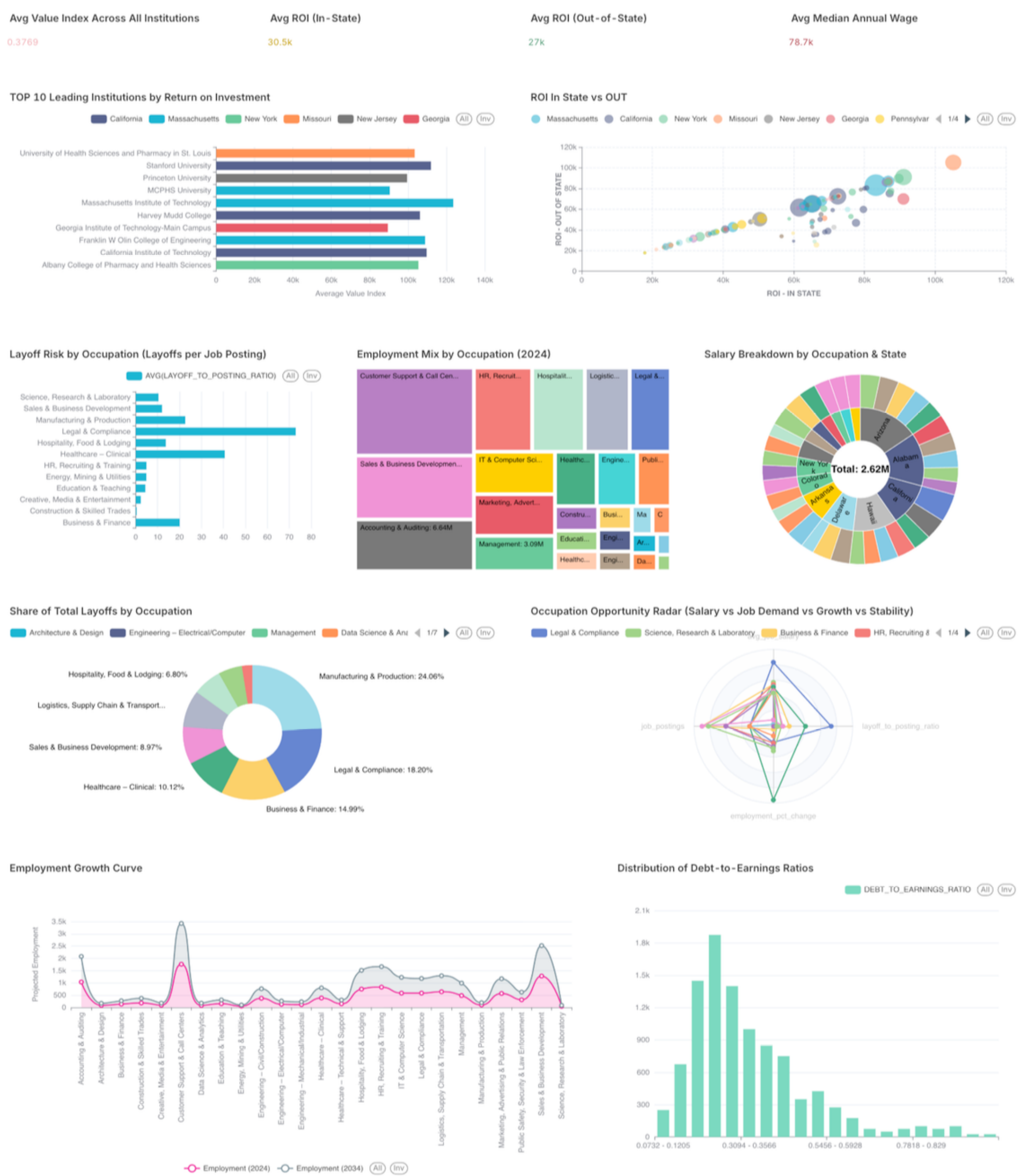
D. End to End Data Pipeline (Lineage)

IV. Data Visualization

Apache Superset or a similar BI tool is used for high-level monitoring and visual analysis. Dashboards connect directly to the analytical marts (ANALYTICS schema) to visualize:

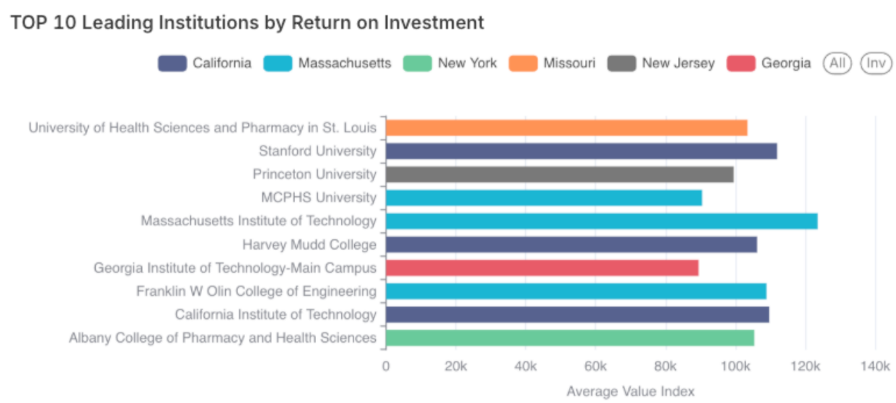
- Real-time job demand and velocity by occupation group.
- Layoff risk heatmaps correlated with industry and degree concentration.
- Aggregate ROI score distribution across institutions and regions.

A. Dashboard:


Avg Value Index Across All Institutions
0.3769
Avg ROI (In-State)
30.5k
Avg ROI (Out-of-State)
27k
Avg Median Annual Wage
78.7k
TOP 10 Leading Institutions by Return on Investment
California
Massachusetts
New York
Missouri
New Jersey
Georgia
University of Health Sciences and Pharmacy in St. Louis
Stanford University
Princeton University
MCPHS University
Massachusetts Institute of Technology
Harvey Mudd College
Georgia Institute of Technology-Main Campus
Franklin W Olin College of Engineering
California Institute of Technology
Albany College of Pharmacy and Health Sciences
Average Value Index
ROI In State vs OUT
ROI - OUT OF STATE
ROI - IN STATE
Layoff Risk by Occupation (Layoffs per Job Posting)
AVG(LAYOFF_TO_POSTING_RATIO)
Science, Research & Laboratory
Sales & Business Development
Manufacturing & Production
Legal & Compliance
Hospitality, Food & Lodging
Healthcare – Clinical
HR, Recruiting & Training
Energy, Mining & Utilities
Education & Teaching
Creative, Media & Entertainment
Construction & Skilled Trades
Business & Finance
Employment Mix by Occupation (2024)
Customer Support & Call Cen...
Sales & Business Developmen...
Accounting & Auditing: 6.64M
IT & Computer Sci...
Marketing, Advert...
Management: 3.09M
Salary Breakdown by Occupation & State
Total: 2.62M
Share of Total Layoffs by Occupation
Architecture & Design
Engineering – Electrical/Computer
Management
Hospitality, Food & Lodging: 6.80%
Logistics, Supply Chain & Transport...
Sales & Business Development: 8.97%
Healthcare – Clinical: 10.12%
Manufacturing & Production: 24.06%
Legal & Compliance: 18.20%
Business & Finance: 14.99%
Occupation Opportunity Radar (Salary vs Job Demand vs Growth vs Stability)
Legal & Compliance
Science, Research & Laboratory
Business & Finance
job_postings
layoff_to_posting_ratio
employment_pct_change
Employment Growth Curve
Projected Employment
Employment (2024)
Employment (2034)
Distribution of Debt-to-Earnings Ratios
DEBT_TO_EARNINGS_RATIO
0.0732 - 0.1205
0.3094 - 0.3566
0.5456 - 0.5928
0.7818 - 0.829

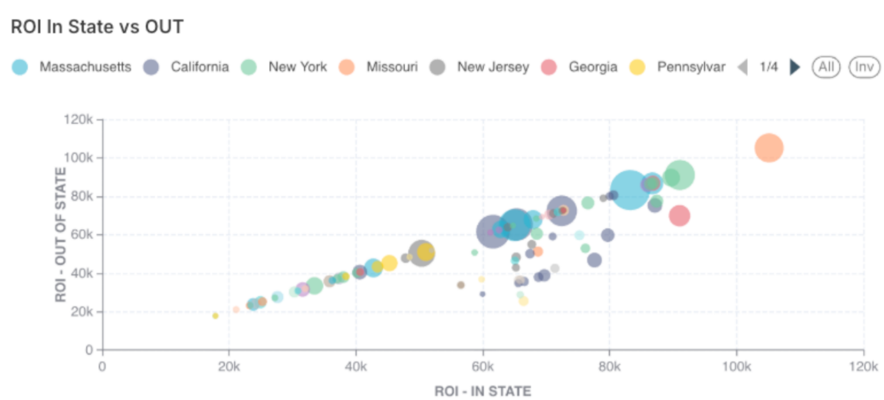
1. TOP 10 Leading Institutions by ROI

A horizontal bar chart identifying the "gold standard" schools. It lists high-performers like **MIT, Princeton, Georgia Tech, Harvey Mudd, and Caltech**. These schools consistently produce the highest earnings relative to their debt/cost.



2. ROI In State vs. OUT (by State)

A grouped bar chart comparing the financial benefit for residents vs. non-residents. States like **California, Massachusetts, and New York** are highlighted, showing that while out-of-state tuition is higher, the ROI often remains high due to the strong local job markets.

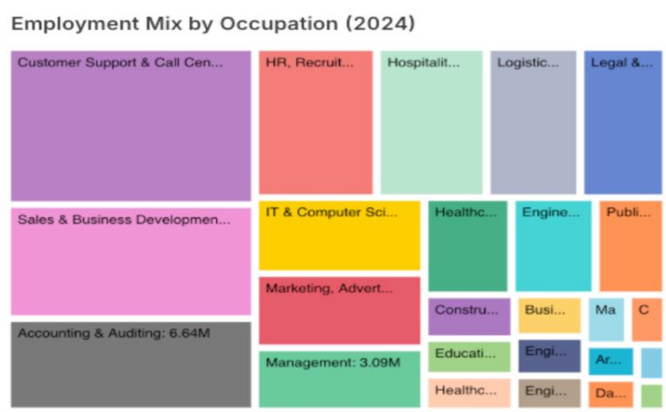


3. Layoff Risk by Occupation (Ratio Chart)

This is the visual representation of your **Layoff Risk Index**. It shows the ratio of layoffs per job posting. A high bar here is a "Warning Signal" that an industry is volatile, even if salaries are high.

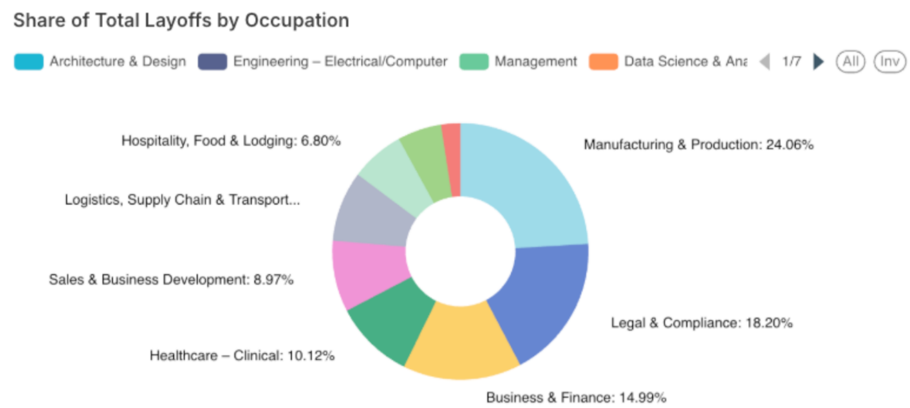
4. Employment Mix by Occupation (2024)

A treemap showing the "density" of the workforce. Large blocks indicate massive employment sectors (like Business & Finance), while smaller blocks represent niche fields.



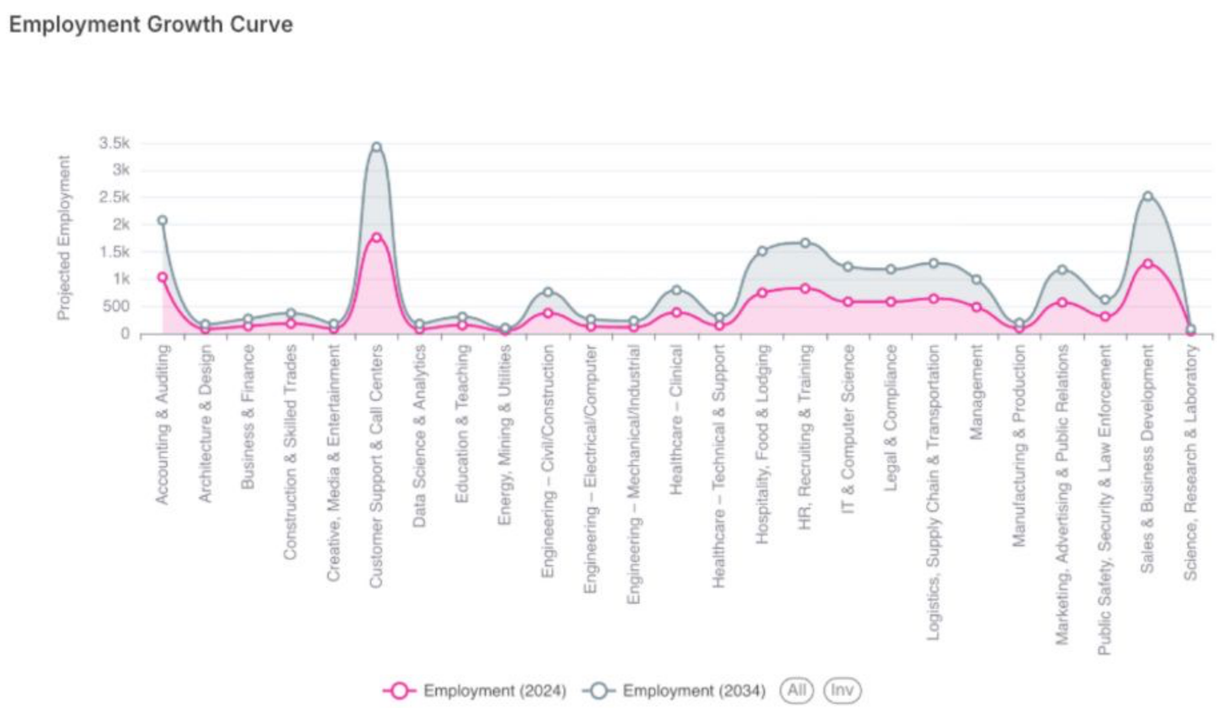
5. Share of Total Layoffs by Occupation (Pie Chart)

This breaks down where job cuts are happening most. In your data, **Manufacturing & Production (24.0%)** and **Legal & Compliance (18.2%)** show significant shares of total layoffs, signaling higher volatility in those sectors.



6. Employment Growth Curve

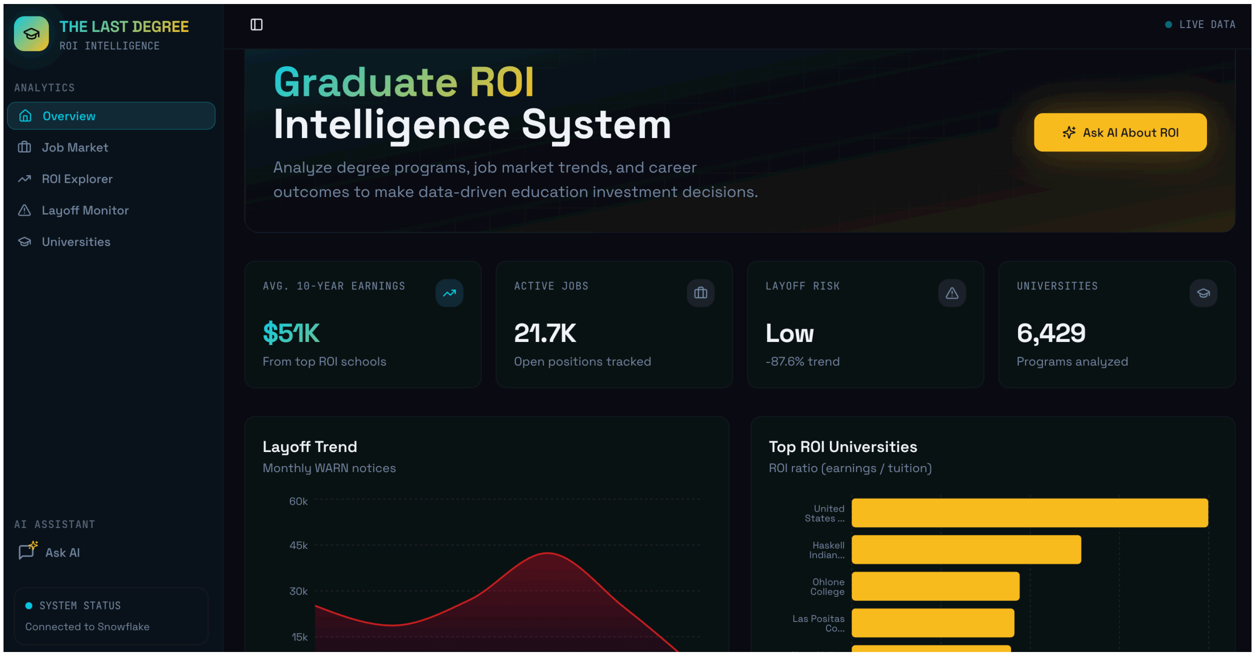
This area/line chart visualizes the **BLS Projections**. It shows which career paths are expected to expand (e.g., Healthcare, Data Science) and which are stagnating over the next decade.



B. Frontend Application

The user interface for this intelligence system is the "**The Last Degree**" web application, built using React. This application consumes the data materialized in the all the tables stored in raw schema and runs SQL query generated by AI agent in real time to deliver personalized, actionable insights.

The application's core feature is the **SQL AI Agent**, which allows non-technical users to query the underlying data marts directly using natural language, enabling real-time, sophisticated analysis.



Link : https://thelastdegree.dev/

V. CONCLUSION AND FUTURE WORK

A. Core Message: Navigating the Future of Work

"The Last Degree" successfully delivers on its promise to provide clarity in an increasingly volatile educational and professional landscape. By architecting a system that seamlessly integrates historical institutional outcomes with real-time labor market signals, we have turned market ambiguity into a strategic advantage. This system stands as an essential intelligence platform, empowering students, educators, and policymakers to navigate the future of work with data-driven confidence rather than speculation.

B. Summary of Technical Success

The project effectively demonstrated the power of a modern data stack in solving complex, multi-domain problems:

- **Automated Intelligence:** Four distinct data pipelines—College Scorecard, Adzuna, BLS, and WARN were fully automated via programmatic API ingestion.
- **The Hybrid Advantage:** Our **ETL/ELT hybrid model** proved that separating structural preparation (Airflow) from analytical logic (dbt/Snowflake) creates a scalable and auditable "Single Source of Truth."
- **Risk-Adjusted Decision Making:** By implementing the **Layoff Risk Index**, we moved beyond traditional ROI metrics to provide a multidimensional view of career stability.

C. Future Vision

As the labor market continues to shift due to technological automation, the roadmap for "The Last Degree" focuses on three strategic pillars:

- **Enhanced Personalization:** We aim to evolve the existing SQL AI Agent into a comprehensive **Conversational Analytics Partner**. This assistant will go beyond data retrieval to offer personalized career path recommendations based on a user's specific skill set, financial goals, and personal risk appetite.
- **Deeper Skill Analysis:** Future iterations will integrate new, granular data sources to analyze the "atomic" level of the workforce measuring **skill supply vs. demand** at a hyper-local level to identify emerging specialization opportunities.
- **Global Expansion:** While the current framework is rooted in domestic data, the architecture is designed for scale. We seek to incorporate **international education and labor market data** to provide a unified, worldwide perspective on the global return on education.

GitHub Link (Backend): The-Graduate-ROI-Intelligence-System

GitHub Link (Frontend): ai-data-insights

REFERENCES

[1] U.S. Department of Education. College Scorecard. https://collegescorecard.ed.gov/.
[2] Adzuna. Job market insights and labor market analytics. https://www.adzuna.com/.
[3] WARN Act notices. https://edd.ca.gov/en/jobs_
[4] U.S. Bureau of Labor Statistics. https://www.bls.gov.
[5] Snowflake Developer Guide. https://docs.snowflake.com/en/developer.
[6] Apache Airflow. https://airflow.apache.org/docs/.